
Lecture 12: Molecular Transitions

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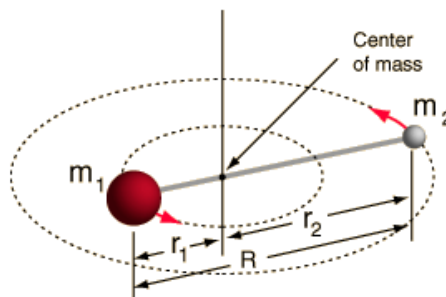
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Molecular Transitions

- Rotational Transitions (mostly in Radio)
 - Linear Molecules (e.g, Diatomic Molecules)
 - Energy Levels and Selection Rules
 - Radiative Transfer and Line Intensities
 - Partition Function and Column Density
 - NonLinear Molecules
 - Symmetric top Molecules
 - Spherical top Molecules
 - Asymmetric top Molecules
- Vibrational Transitions (in Infrared)

Diatomic Molecules: Linear

For examples: H₂, CO, CS, SiO ...



$$R = r_1 + r_2 \qquad m_1 r_1 = m_2 r_2$$

$$I = m_1 r_1^2 + m_2 r_2^2 = \frac{m_1 m_2}{m_1 + m_2} (r_1 + r_2)^2 = \mu R^2 \qquad \text{with } \mu = \frac{m_1 m_2}{m_1 + m_2}$$

Then the **rotational energy** is

$$E = \frac{I\omega^2}{2} = \frac{L^2}{2I} \qquad \text{with angular momentum } L = I\omega$$

Due to **quantization**, we have

$$|L^2| = \hbar^2 J(J+1) \qquad \text{with } J = 0, 1, 2, \dots$$

thus,

$$\Rightarrow E_J = \frac{\hbar^2 J(J+1)}{2I} = BJ(J+1) \qquad \text{with } B = \frac{\hbar^2}{2I}$$

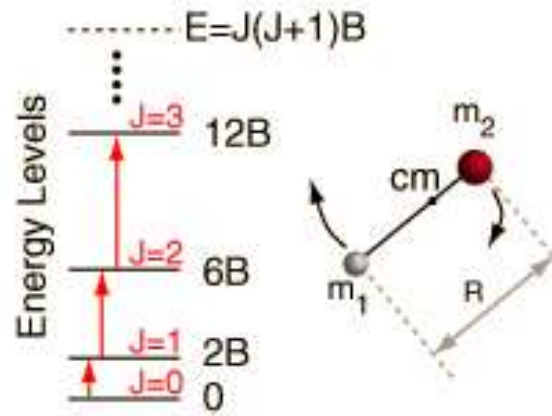
Diatomic Molecules

For examples: H₂, CO, CS, SiO ...

The energy levels:

$$E_J = \frac{\hbar^2 J(J+1)}{2I} = BJ(J+1) \quad (\text{ergs}) \quad \text{with} \quad B = \frac{\hbar^2}{2I}$$

Selection Rule: $\Delta J = \pm 1$.



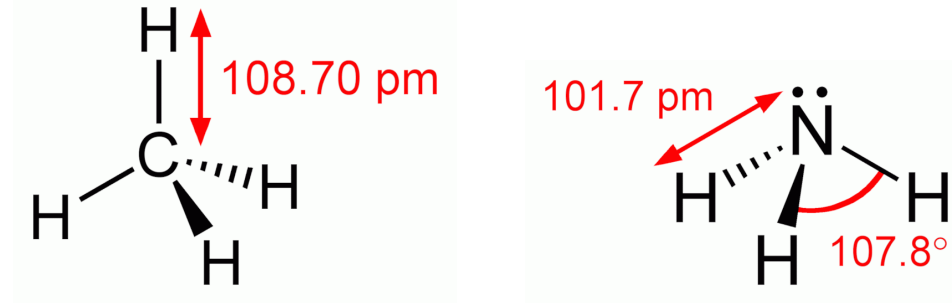
Transition $J + 1 \rightarrow J \Rightarrow$ Emission line with energy:

$$\Delta E = E_{J+1} - E_J = B(J+1)(J+2) - BJ(J+1) = 2B(J+1)$$

and frequency $\nu = \frac{\Delta E}{h} = 2\frac{B}{h}(J+1)$ and $\nu_{10}(J=1 \rightarrow 0) = 2B/h \Rightarrow \nu = \nu_{10}(J+1)$.

Symmetric Top Molecules

For examples: CH₄ (Methane), NH₃ (Ammonia), CH₃CN (Methyl Cyanide), ...



The energy levels:

$$E = \frac{I_x \omega_x^2}{2} + \frac{I_y \omega_y^2}{2} + \frac{I_z \omega_z^2}{2} = \frac{L_x^2}{2I_x} + \frac{L_y^2}{2I_y} + \frac{L_z^2}{2I_z}$$

Symmetric Top, two moments of inertia are the same: (1) Prolate (football): $I_x > I_y = I_z$ (CH₄, NH₃), (2) Oblate (flying saucer): $I_x = I_y > I_z$ (CH₃CN). In General,

$$I_x = I_y \equiv I_B \qquad I_z \equiv I_C$$

With $L^2 = L_x^2 + L_y^2 + L_z^2$, we have

$$E = \frac{L^2}{2I_B} + L_z^2 \left(\frac{1}{2I_C} - \frac{1}{2I_B} \right)$$

Symmetric Top Molecules

For examples: CH₄ (Methane), NH₃ (Ammonia), CH₃CN (Methyl Cyanide), ...

The energy levels:

$$E = \frac{L^2}{2I_B} + L_z^2 \left(\frac{1}{2I_C} - \frac{1}{2I_B} \right)$$

Because of quantization, we define

$$|L^2| = \hbar^2 J(J+1) \quad |L_z^2| = \hbar^2 K^2 \quad \text{with } K = -J, \dots, 0, \dots, J$$

then

$$E_{J,K} = J(J+1) \frac{\hbar^2}{2I_B} + K^2 \left(\frac{\hbar^2}{2I_C} - \frac{\hbar^2}{2I_B} \right)$$

Define

$$B = \frac{\hbar^2}{2I_B} \quad C = \frac{\hbar^2}{2I_C}$$

then

$$E_{J,K} = BJ(J+1) + K^2(C - B)$$

Selection Rule:

$$\Delta J = \pm 1 \quad \Delta K = 0$$

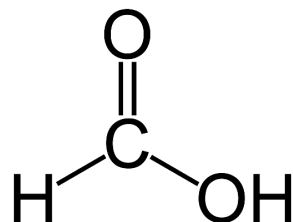
Spherical Top and Linear Molecules

A spherical top molecule, can be considered as a special case of symmetric tops with equal moment of inertia about all three axes ($I_x = I_y = I_z$). E.g., carbon tetrachloride (CCl_4), nitrogen tetrahydride (NH_4), ammonium ion (NH_4^+). Unlike other molecules, spherical top molecules have no net dipole moment, and hence they do not exhibit a pure rotational spectrum.

Linear molecules (diatomic molecules, also carbon dioxide ($\text{O}=\text{C}=\text{O}$), hydrogen cyanide (HCN), carbonyl sulfide ($\text{O}=\text{C}=\text{S}$), chloroethyne (HCCl), acetylene (HCCH) etc): A linear molecule is a special case in this regard. These molecules are cylindrically symmetric and one of the moment of inertia (I_x , which is the moment of inertia for a rotation taking place along the axis of the molecule) is negligible (i.e. $I_x \ll I_y = I_z$).

Asymmetric Top Molecules

For examples: HCOOH (Formic acid), C₂H₅CN (ethyl cyanide)



The energy levels:

$$E = \frac{I_x \omega_x^2}{2} + \frac{I_y \omega_y^2}{2} + \frac{I_z \omega_z^2}{2} = \frac{L_x^2}{2I_x} + \frac{L_y^2}{2I_y} + \frac{L_z^2}{2I_z}$$

Asymmetric Top:

$$I_x \neq I_y \neq I_z$$

Then

$$E = \frac{AL_x^2}{\hbar^2} + \frac{BL_y^2}{\hbar^2} + \frac{CL_z^2}{\hbar^2} \Rightarrow E = E_{J,K_-,K_+}$$

Selection rule: $\Delta J = 0, \pm 1$.

Applications

■ as Probes of Kinematics & Masses

- line velocity

- line width, virial mass (i.e. escape velocity vs. grav. energy => mass)

For a stable, self-gravitating, spherical distribution of equal mass objects (stars, galaxies), the total kinetic energy of the objects is equal to minus 1/2 times the total grav. potential energy.

$$\frac{1}{2} M v^2 = -\frac{1}{2} \left(-\frac{1}{2} \frac{G M^2}{R} \right) \Rightarrow M = 2 \frac{R v^2}{G}$$

where R is the size, v^2 is the mean of the squares of v i.e, linewidth.

■ as Probes of Temperature

- brightness temperature in optically thin lines

- line intensity ratios, particularly, K-ladder for symmetric tops

■ as Probes of Density

- thermalized emission

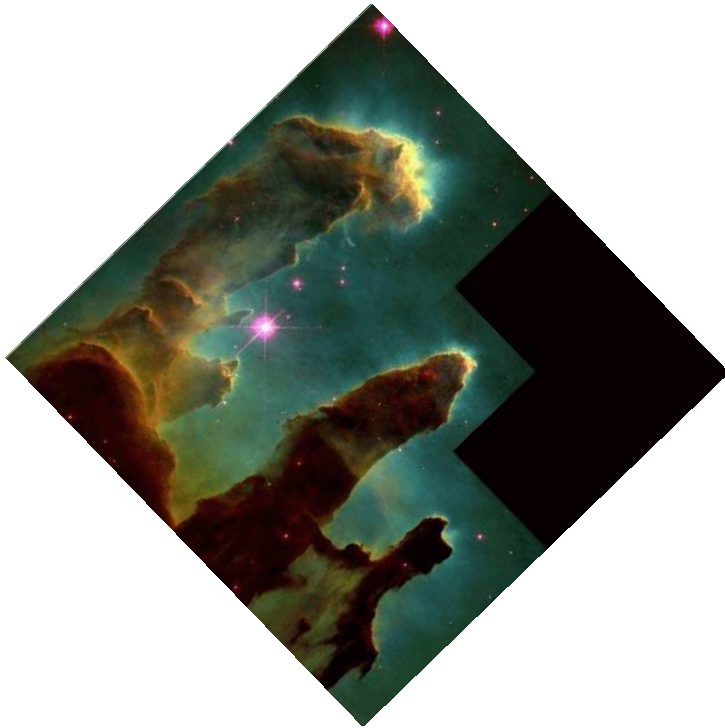
■ as Probes of Molecular Chemistry itself

- abundances

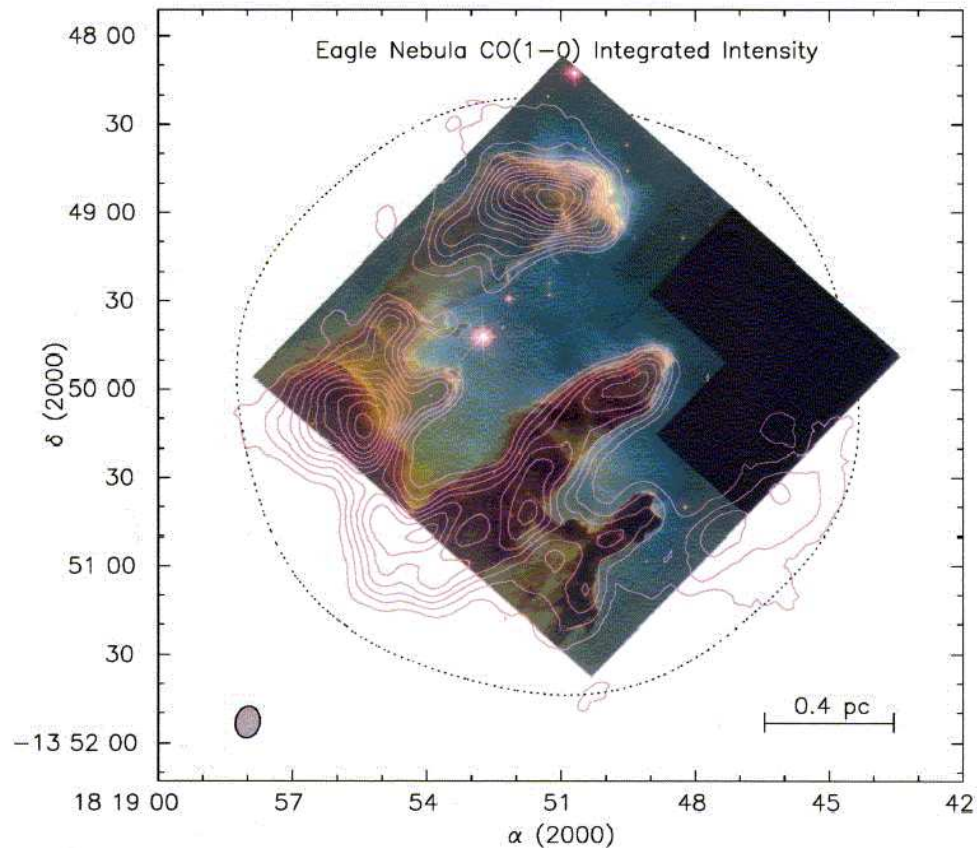
M16-Eagle Nebula

Pillars of Creation!

(Left) (courtesy HST/STSCI). The HST has given us breathtaking views of the nebular sites of star formation on our galaxy. The gaseous pillars of the Eagle nebula shown above are spectacular. (Right) An infrared view of the pillars reveals different structures (ESO).

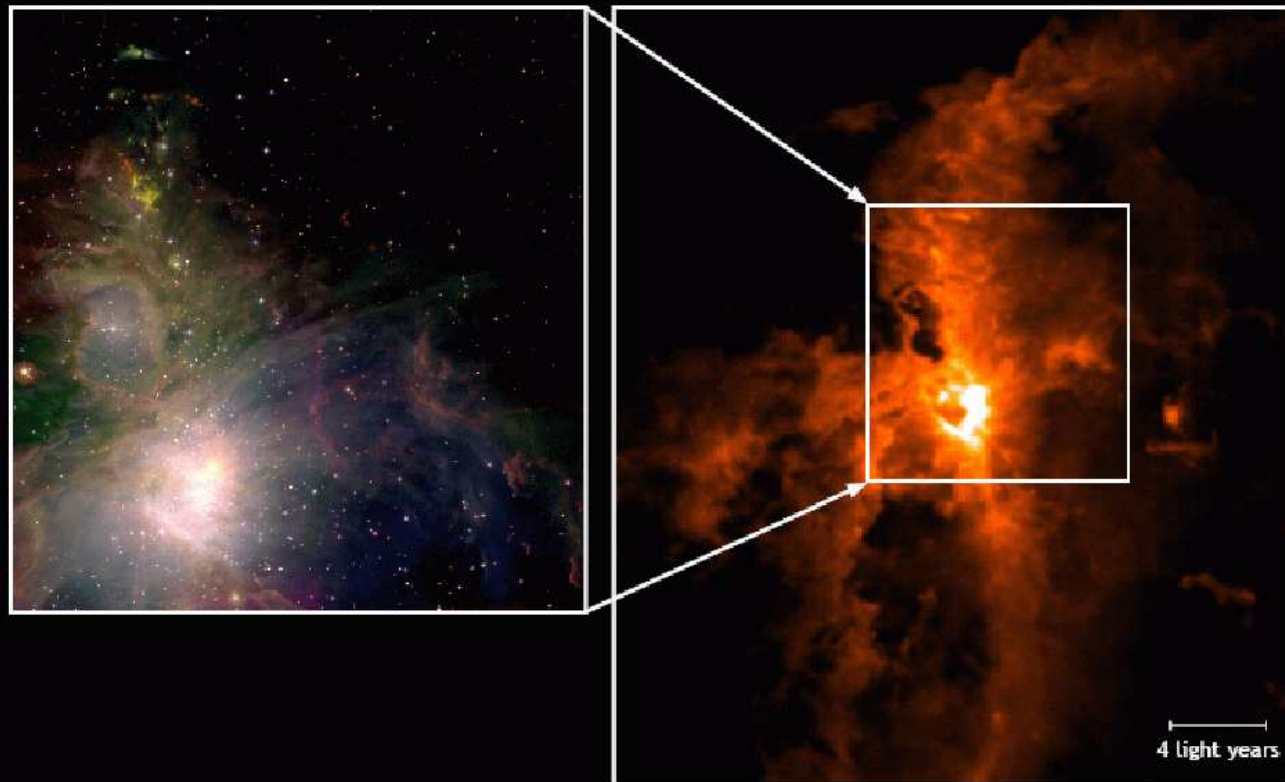


M16-Eagle Nebula



CO Map from BIMA (Pound 1998): Velocity gradients in the molecular gas reach up to $20 \text{ km s}^{-1} \text{ pc}^{-1}$ – enormous in comparison to typical clouds – with position angles that point directly back to the O stars. $\Rightarrow$ Molecular cloud being **eaten away**, sculpted by a photoionizing gale from nearby O stars. Large "elephant trunks" take shape as denser regions in the cloud take longer to photoevaporate and shield material behind them. Dozens of "evaporating gas globules" (EGGs) are tiny versions of the elephant trunks that were once dense cores deep inside the molecular cloud.

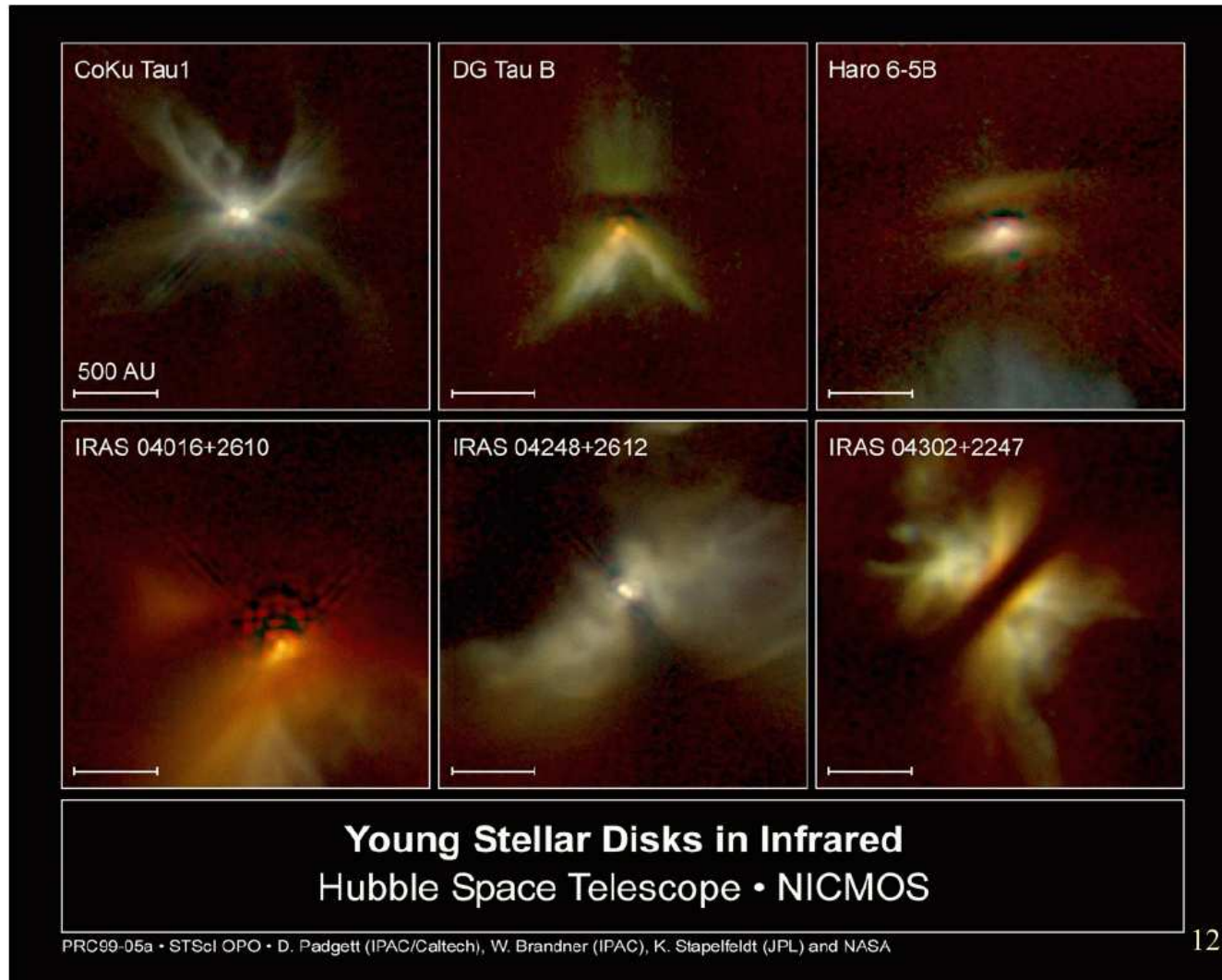
A Look of Orion Nebula in CO



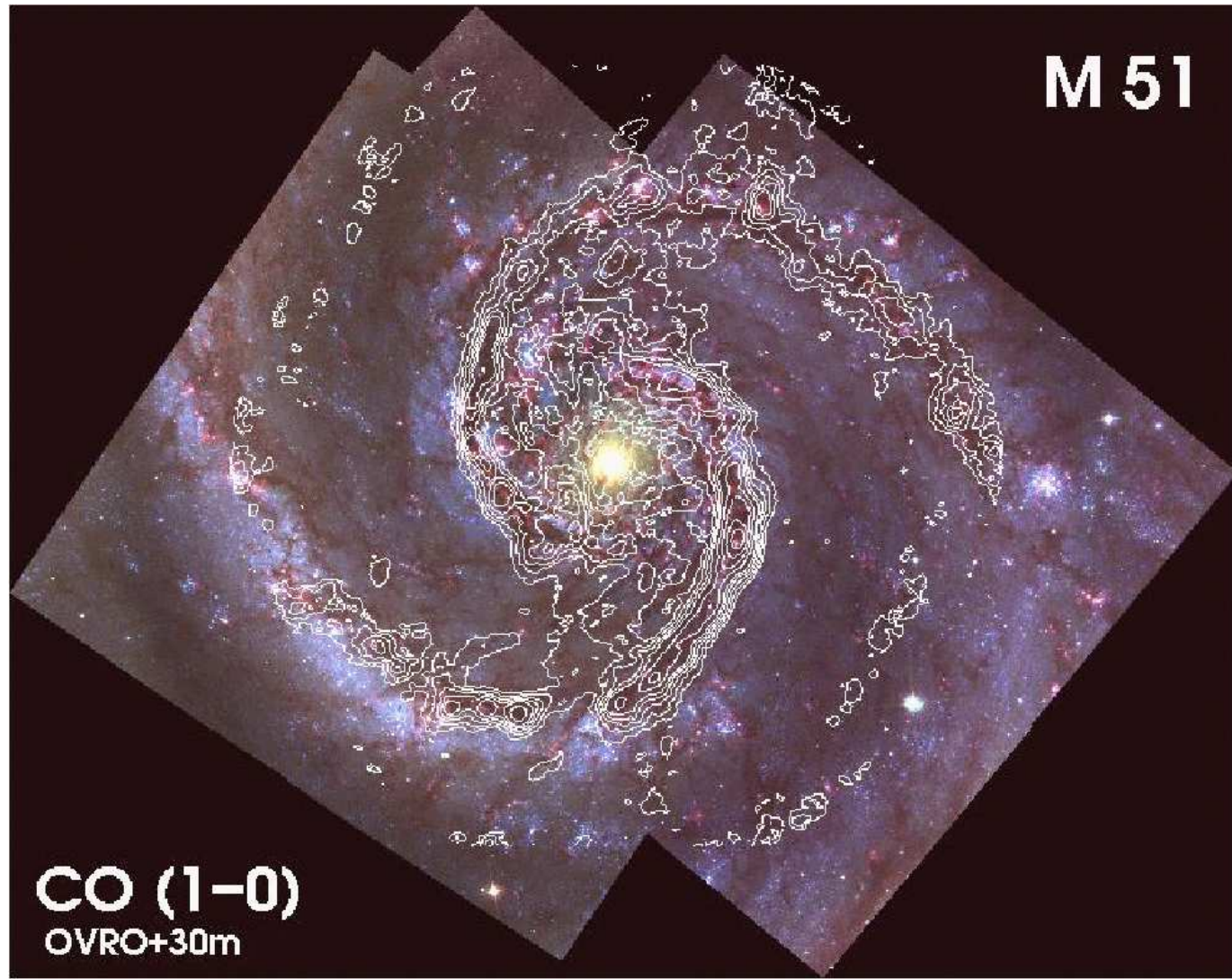
Slicing the Universe with HARP/ACSIS - A New Look at Orion

The image on the right shows Orion obtained with HARP/ACSIS on the James Clerk Maxwell Telescope (JCMT) in the submillimetre wavelength, while the image on the left shows a comparison image obtained with WFCAM on the United Kingdom Infrared Telescope (UKIRT) in the infrared wavelengths. The JCMT submillimetre images shows the dense cloud material from which new stars form. The UKIRT infrared image shows the young stars and the heated gas filaments surrounding them. Both telescopes are operated by the Joint Astronomy Centre in Hilo, Hawai'i. The bar corresponds to four light years, which is equal to the distance from our Sun to the nearest star. (JAC-PR-07-01)

YSO disks

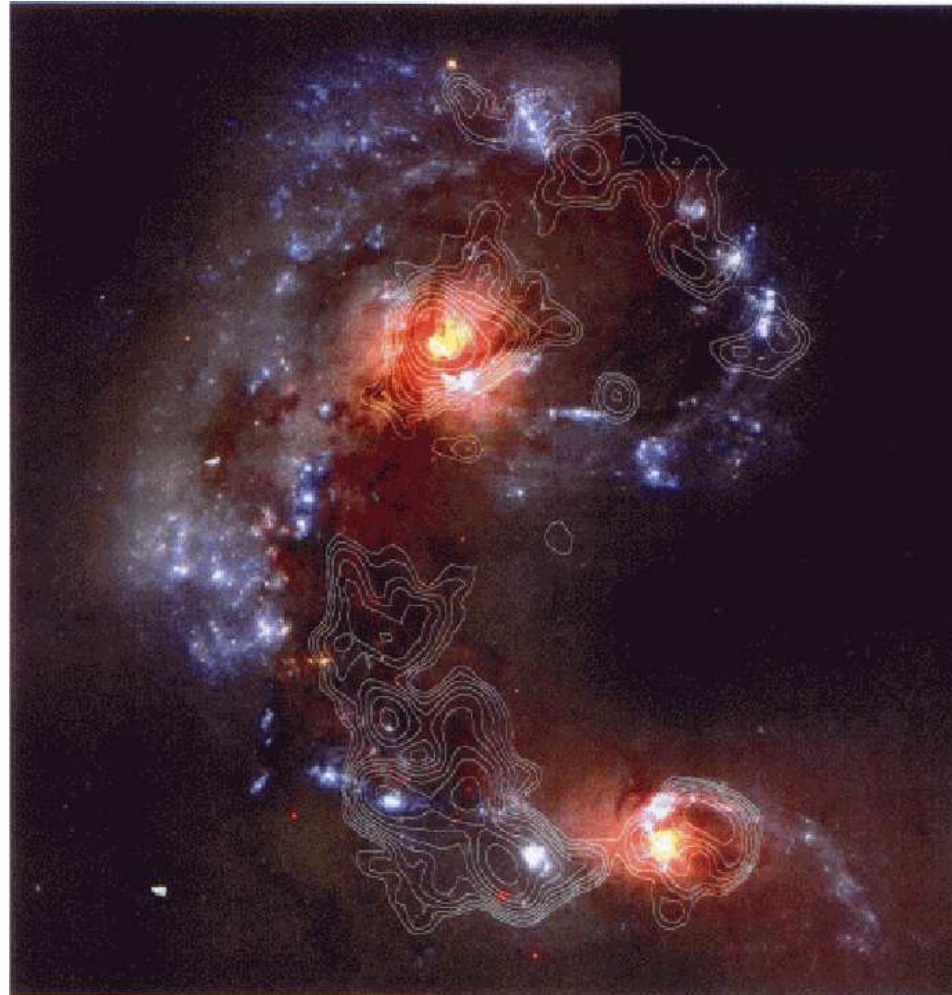


CO Gas in M51



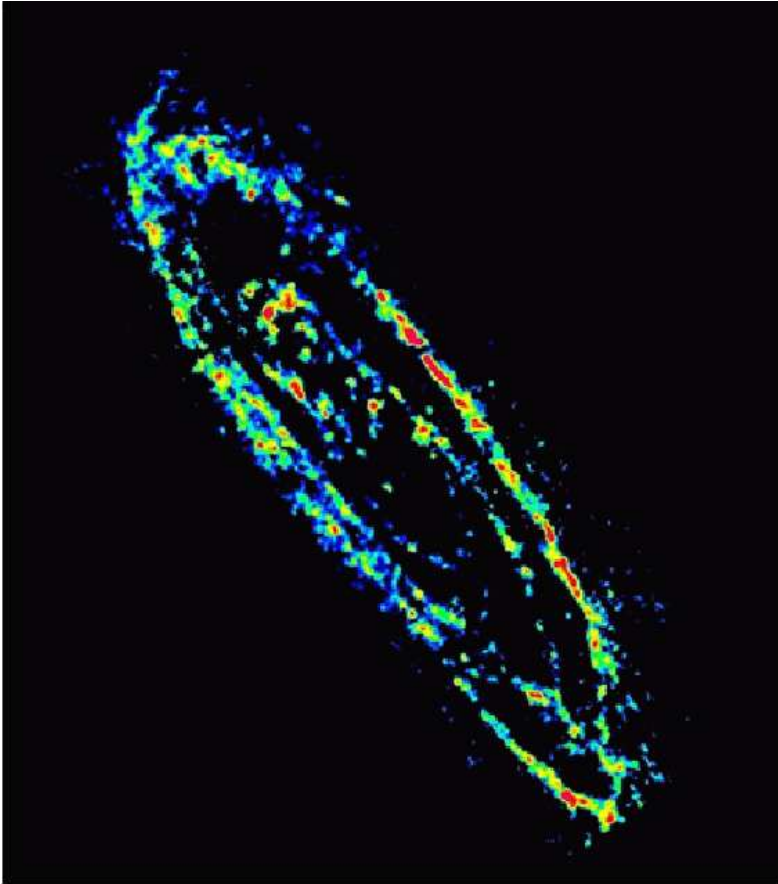
⇒ Dark is not empty. Star forming in the dusty/gaseous spiral arms!

CO Gas in Antennae



Contours of CO(1-1) emission in two colliding galaxies, NGC4038/4039 (the Antennae) overlaid on an HST image. Massive concentrations of molecular gas are located at the nucleus of each galaxy and in the interaction region.

CO Gas in M31

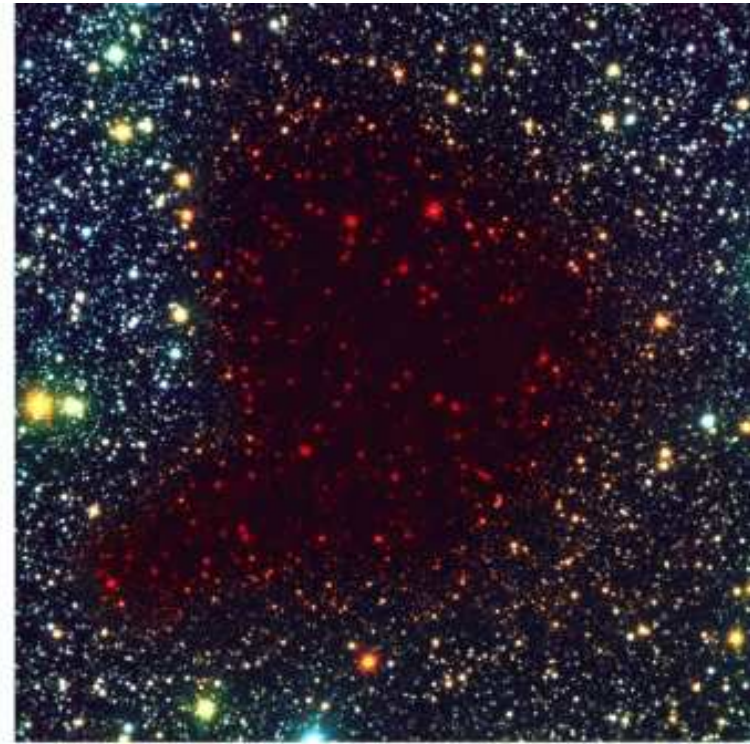


CO and HST images of M31.

Dark Cloud B68



B, V, I



B, I, K

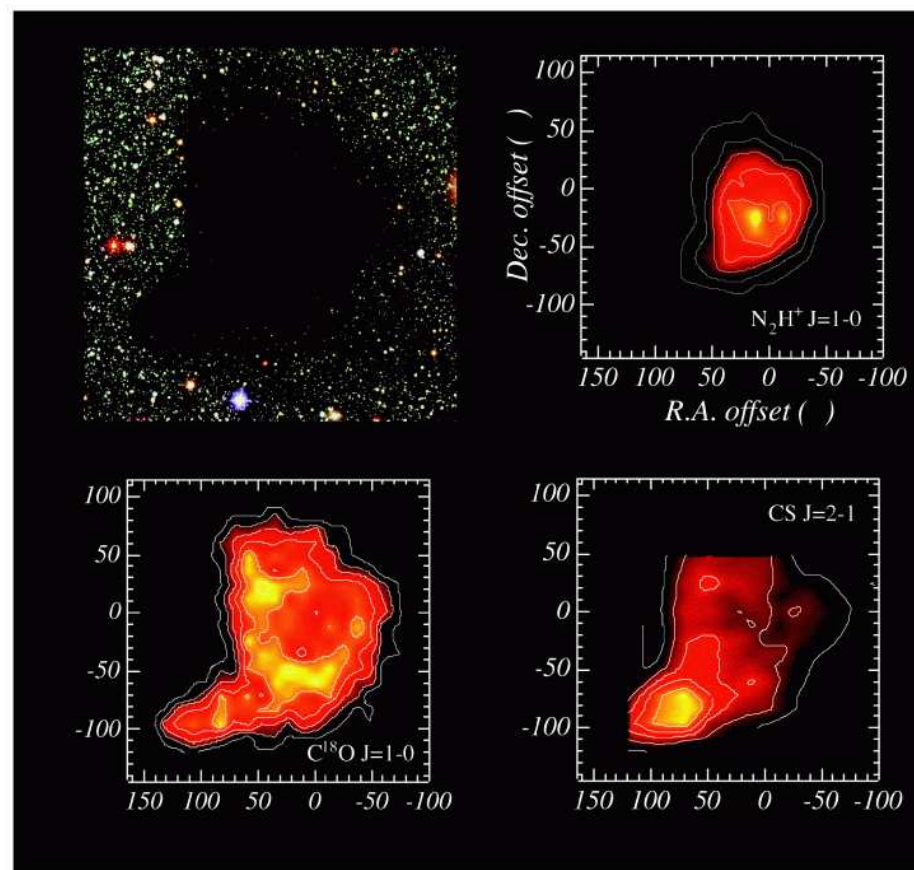
Pre-Collapse Black Cloud B68 (comparison)
(VLT ANTU + FORS 1 - NTT + SOFI)

ESO PR Photo 02c/01 (10 January 2001)

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Dark Cloud B68



(Lada et al 2003, ApJ 586, 286). Deep optical image of the dark globule B68 (Alves et al. 2001) along with contour maps of integrated intensity from molecular emission lines of N_2H^+ , C^{18}O , and CS. Differences in appearance of the molecular line maps are primarily due to the effect of increasing molecular depletion onto grains, with N_2H^+ being the least depleted and CS the most depleted of these species in the cloud core (Bergin et al. 2002). Low Temperature because of shielding?

Polarized Line Radiation

Measuring magnetic fields - molecular line polarization

- Zeeman effect

- Circular polarization

- (Dutch) Zeeman

When Zeeman examined the dark spectral lines he found three where there should have been just one—the magnetic field had split the lines into triplets.

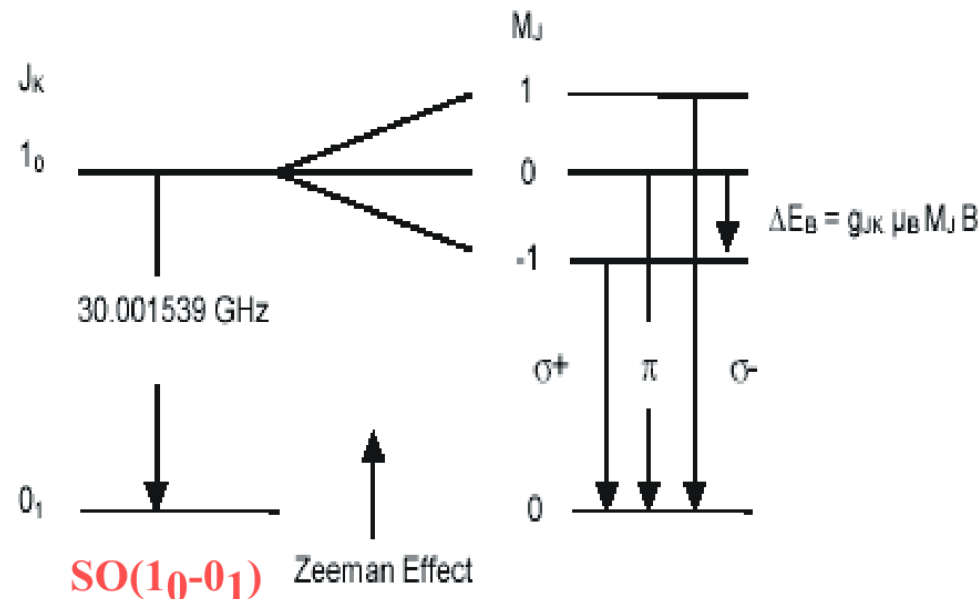
This was an important discovery, for it proved Lorentz's much earlier hypothesis that the atom is comprised of charged particles. Since electrons (which emit light) are charged, they are easily affected by a strong magnetic field.

- line splitting

- strength of B-field along the line of sight

Zeeman effect

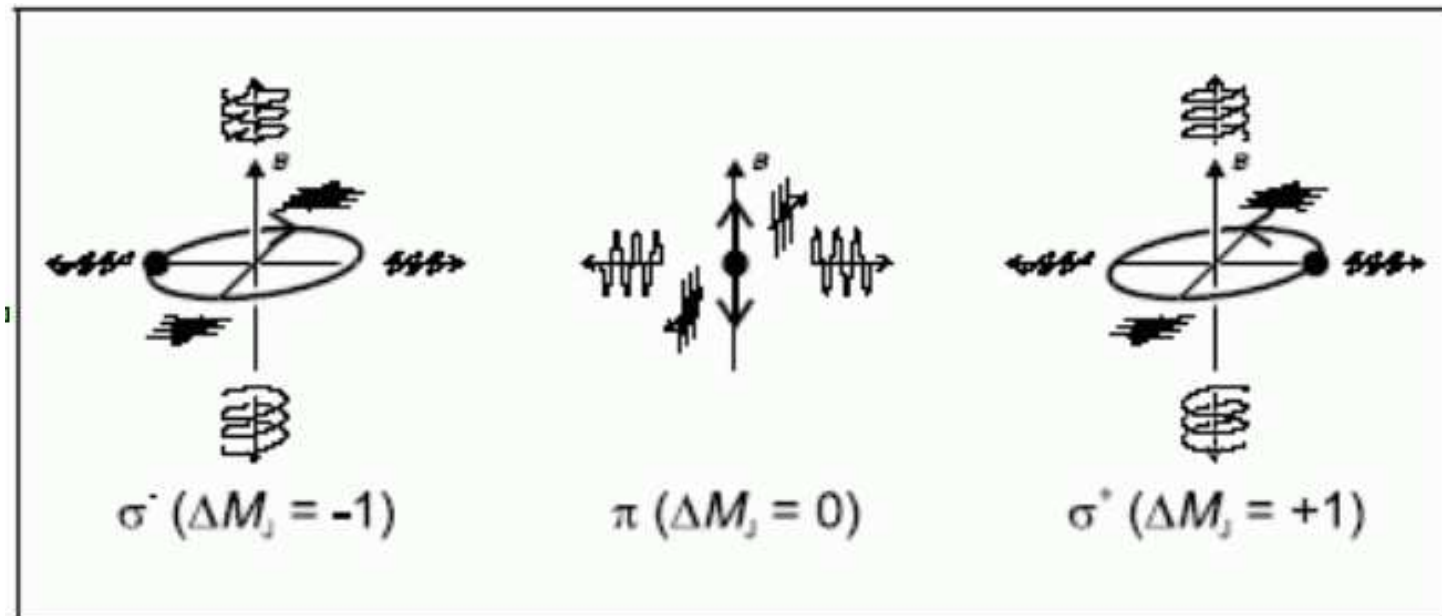
- Splitting of spectral lines into multiple components due to the coupling of an atom's or molecule's magnetic moment with an external magnetic field. $\sigma+$ and $\sigma-$ has signal with different circular polarization. Different g factor for different molecules and transition. μ_B is the projection of the magnetic dipole moment of the electron onto the direction of the field. M_j is the magnetic quantum number of the electron, e.g., +1, 0, -1.



- Suitable atoms or molecules are HI, OH, H_2O , CSS, CN, SO, etc. Most successful detection cases of Zeeman measurement are with HI, OH, H_2O .

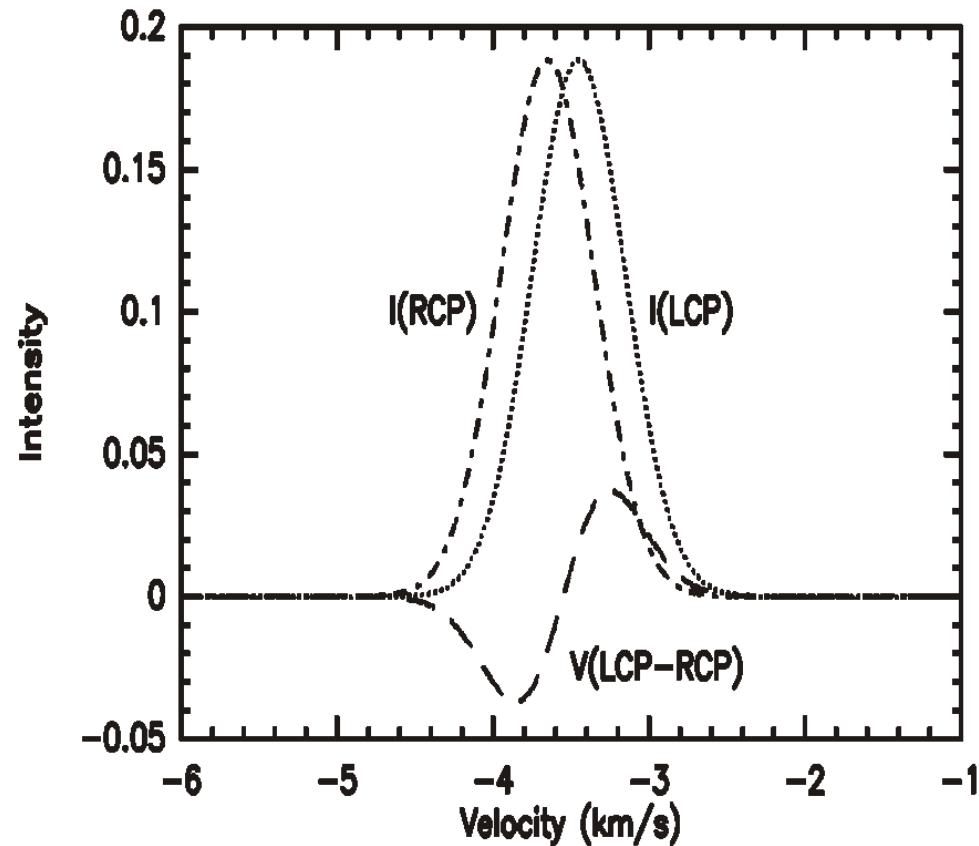
Zeeman effect

In classical terms, the case $\Delta M_J = 0$ corresponds to an infinitesimal dipole oscillating parallel to the magnetic field. No quanta are emitted in the direction of the magnetic field. The light emitted perpendicular to the magnetic field is linearly polarized, whereby the E-vector oscillates in the direction of the dipole and parallel to the magnetic field. Conversely, in the case $\Delta M_J = \pm 1$, most of the quanta travel in the direction of the magnetic field. In classical terms, this case corresponds to two parallel dipoles oscillating with a phase difference of 90° . The super positioning of the two dipoles produces a circulating current. Thus, in the direction of the magnetic field, circularly polarized light is emitted; in the positive direction, it is clockwise circular for $\Delta M_J = +1$ and counter clockwise circular for $\Delta M_J = -1$.

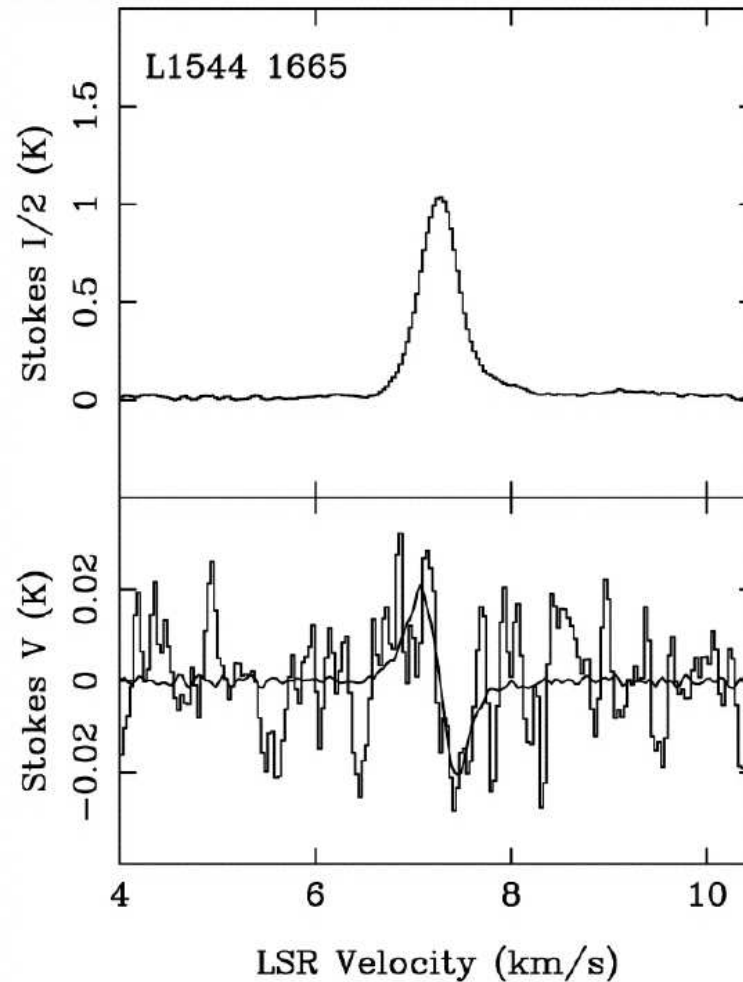


Zeeman effect

- If the splitting is large (RCP, LCP well separated), the total field strength can be derived (only OH and H₂O maser)
- If the splitting is small, it provides only the "line of sight" component B_{los} .



Zeeman effect



B (l.o.s.) $\sim 10\mu\text{G}$. (Crutcher and Troland 2000 ApJL.) For circularly polarized emission, Stokes $Q = U = 0$, $I = S$ and $V = \pm S$.

Zeeman effect

- Bear in mind that in most cases, only line-of-sight or plane-of-sky component is measured.
- Observational data supporting the existing star formation theories are still rare, because it is hard to measure the strength of interstellar magnetic fields.

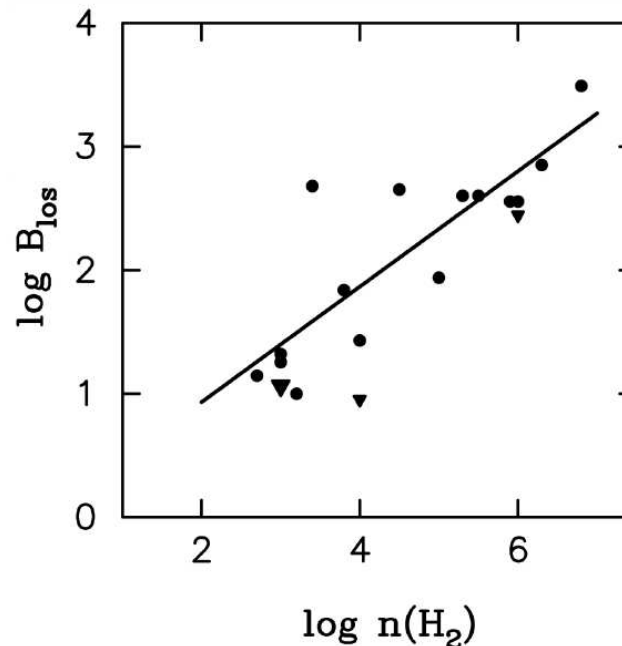
Theories:

The B- ρ relation: $|B| \propto \rho^k$, $k = 1/2$ to $1/3$ (Mouschovias, 1985).

Observations:

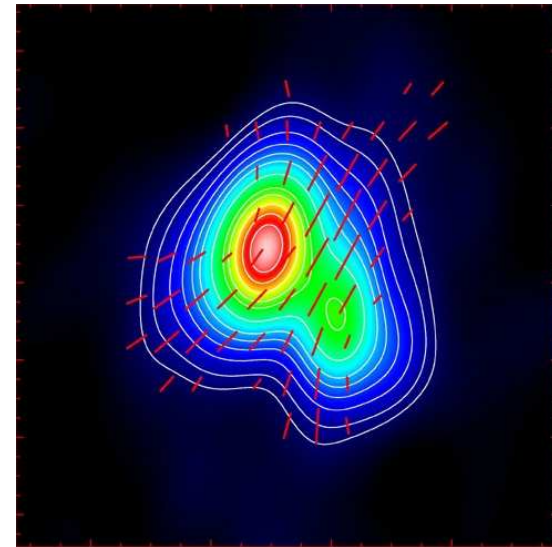
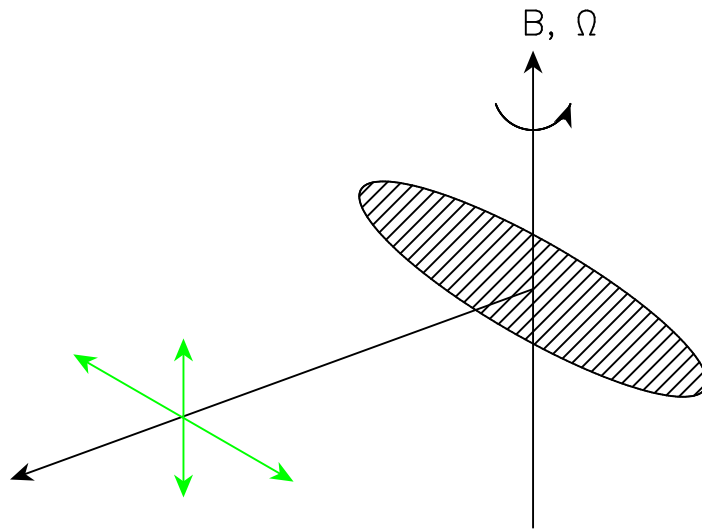
Crutcher (1999) reviewed 15 detections and 3 upper limits towards the dense region.

$$B_{\text{los}} \propto \rho^{0.47 \pm 0.08}$$



Dust Polarization

If the dust grains are **non-spherical**, and some process aligns the grains, one would expect the thermal emission to be polarized. Eg., if the grains have an electric dipole and as more dipoles are created along the largest axis of the grains, they are aligned by a B field (via torque) with the largest axis aligned perpendicular to the B field, then the thermal emission will be more intense perpendicular to the B field $\Rightarrow$ **measuring the direction of B field**.



SMA has done some really nice dust polarization observations toward star forming cores.